



TVS CREDIT E.P.I.C 8.0 // GRAND FINALE // PROBLEM STATEMENT (c)

# SMART LENDING DECISION HUB

Autonomous Geospatial AI, Sentinel-2 Crop Telemetry & Harvest-Aligned Dynamic Repayments for Rural Bharat

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< 48H SANCTIONS

10M SATELLITE NDVI

TREESHAP EXPLAINABILITY

HARVEST BULLET EMIS

GEMINI AI COPILOT

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# THE RURAL CREDIT CHASM: EXCLUSION, FRICTION & DEFAULT TRAPS

## 01 // 85% THIN-FILE EXCLUSION

- Over 120M Indian smallholders operate in cash with zero formal CIBIL scores.
- Traditional banks demand 3-year ITRs & audited pay-slips they do not have.
- Forces farmers into predatory moneylenders charging 36% - 60% annualized APR.

## 02 // ₹3,500+ MANUAL AUDIT COST

- Manual field verification takes 14 to 21 days with revenue-patwari site audits.
- Physical inspection costs ₹3,500 - ₹5,000 per application, inflating operational CAC.
- High vulnerability to boundary falsification, tenant disputes, and ghost-plot fraud.

## 03 // RIGID FLAT MONTHLY EMIs

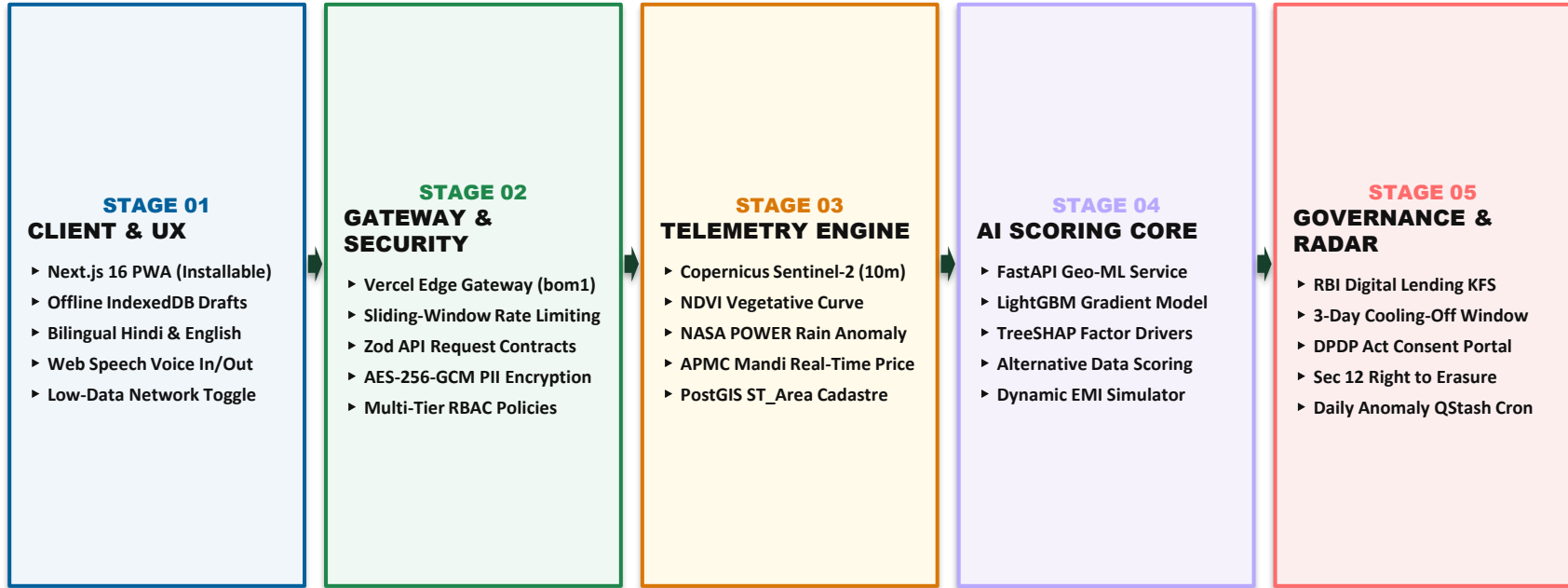
- Conventional banking forces equal monthly EMIs (e.g. ₹12,500/mo) throughout the year.
- During sowing seasons (Jun-Aug), farmer cash is depleted by seeds/fertilizers.
- Triggers structural loan defaults despite guaranteed bumper harvest maturity in 90 days.

## 04 // CLIMATE & MANDI PRICE SHOCKS

- Monsoon deficits (-30% rainfall) and localized pest attacks trigger regional defaults.
- Sudden wholesale APMC Mandi price crashes wipe out expected harvest profits.
- Lenders lack continuous surveillance to restructure loans before accounts turn into NPAs.

**CORE THESIS:** Replace manual friction with objective Sentinel-2 satellite truth, and align debt repayment with agricultural crop cashflows.

# FULL-STACK ARCHITECTURE: 5-PILLAR DECISION BLUEPRINT



## PRODUCTION TECH STACK & INTEGRATIONS:

Next.js 16.3 (Turbopack) • TypeScript 5 • Python 3.11 FastAPI • Supabase Postgres + PostGIS • Upstash Redis/QStash • Google Gemini 2.5 Flash • Copernicus CDSE • NASA POWER • Serwist PWA

# GEOSPATIAL TELEMETRY: SENTINEL-2 NDVI & WEATHER INGESTION

## SATELLITE & CLIMATE INGESTION PIPELINE

### ► 10-Meter Optical Multi-Spectral Resolution:

Ingests Sentinel-2A and 2B constellations revisiting farm parcels every 5 days for real-time cloud-masked telemetry.

### ► Normalized Difference Vegetation Index (NDVI):

Calculated via  $(\text{Band 8 NIR} - \text{Band 4 Red}) / (\text{Band 8 NIR} + \text{Band 4 Red})$ . Direct indicator of active chlorophyll and biomass health.

### ► Cloud Masking (SCL) & SAR Fallback:

Scene Classification Layer masks clouds/shadows; automatic fallback to Sentinel-1 C-Band Synthetic Aperture Radar during monsoon overcast.

### ► NASA POWER Climate Reanalysis:

Ingests 90-day precipitation and computes % deviation against the 10-year district moving average to detect localized drought or flood stress.

### ► PostGIS Cadastral Verification:

Calculates polygon area via `ST_Area(geom::geography)`. Automatically flags discrepancies >20% between declared and measured acreage.

## NDVI UNDERWRITING THRESHOLD MATRIX

### ■ NDVI > 0.65 (OPTIMAL CROP VIGOR)

Condition: High Chlorophyll Biomass

Action: Green Channel Sanction: Up to 90% LTV, preferred 11.5% interest rate, zero physical inspection needed.

### ■ NDVI 0.45 - 0.65 (NORMAL HEALTH)

Condition: Standard Vegetative Growth

Action: Standard Approval: 80% LTV, 12.2% interest rate, automated disbursal directly into farmer's bank account.

### ■ NDVI 0.25 - 0.45 (MOISTURE STRESS)

Condition: Vegetative Growth Deficit

Action: Conditional Approval: Requires borewell/drip irrigation validation or secondary co-applicant guarantee.

### ■ NDVI < 0.25 (BARREN / FAILURE)

Condition: Stunted Crop / Dry Plot

Action: Decline / Advisory: Automatically diverted to PMFBY crop insurance support & soil conditioning advisory.

# EXPLAINABLE AI: LIGHTGBM SCORING & TREESHAP DRIVERS

## THE EXPLAINABILITY IMPERATIVE (WHY SHAP?)

### ► Eliminating the 'Black-Box' Risk:

Traditional deep learning models generate scores without explanation, violating RBI fair-lending mandates and preventing credit officer buy-in.

### ► Local Shapley Additive Explanations:

TreeSHAP calculates exact additive feature contributions for each applicant, showing precisely which variables increased or decreased creditworthiness.

### ► Auditable Credit Memo Generation:

Transforms algorithmic output into a clean, human-readable underwriting sheet with objective risk tier classification (Tier A to D).

### ► Bias & Discrimination Mitigation:

Excludes protected personal attributes; scores strictly on agronomic health, soil potential, weather resiliency, and mandi proximity.

## SHAP WATERFALL DECOMPOSITION (APPLICANT: RAMESH PATIL)

- Base Population Score (Rural Benchmark): 580 / 900
- [+] Sentinel-2 Crop Health (NDVI 0.72 vs 0.52): +85 Points
- [+] Soil Organic Carbon (8.4 g/kg Rich Soil): +42 Points
- [+] Mandi Distance (12 km - Low Transport Friction): +38 Points
- [-] NASA Weather Anomaly (-18% Rainfall Deficit): -25 Points
- [+] Micro-Drip Irrigation (Mitigates Rain Deficit): +30 Points

## FINAL SCORE: 750 / 900 (LOW RISK // TIER A)

Decision: Auto-Approved ₹3,50,000 Tractor Loan

Structure: Harvest-Aligned Bullet Schedule (11.5% APR)

Turnaround: Sanction Letter Generated in < 3 Minutes

# PRODUCT INNOVATION: HARVEST-ALIGNED REPAYMENT LIFECYCLE

## PHASE 1: SOWING

JUN - AUG (3 MO) • Peak Outflow  
(Seeds, Fertilizers, Diesel, Labor)

### ✗ TRADITIONAL FLAT:

EMI: ₹12,500 / mo  
Impact: Severe Deficit → 18.4% Default

### ✓ TVS HARVEST-ALIGNED:

EMI: ₹800 / mo  
Impact: Token Interest → Zero Default

## PHASE 2: VEGETATIVE

SEP (1 MO) • Maintenance Outflow  
(Pesticides & Weeding Sprays)

### ✗ TRADITIONAL FLAT:

EMI: ₹12,500 / mo  
Impact: Liquidity Strain → Missed EMI

### ✓ TVS HARVEST-ALIGNED:

EMI: ₹1,200 / mo  
Impact: Low Maintenance → Cash Intact

## PHASE 3: HARVEST/MANDI

OCT - NOV (2 MO) • Massive Cash Inflow  
(APMC Mandi Wholesale Crop Sale)

### ✗ TRADITIONAL FLAT:

EMI: ₹12,500 / mo  
Impact: Under-collects surplus cash

### ✓ TVS HARVEST-ALIGNED:

EMI: ₹1,15,000 Bullet  
Impact: Syncs with ₹2.5L+ Mandi Sale

## PHASE 4: OFF-SEASON

DEC - MAY (6 MO) • Stabilized Flow  
(Rabi winter crops & dairy income)

### ✗ TRADITIONAL FLAT:

EMI: ₹12,500 / mo  
Impact: Rigid burden on winter sowing

### ✓ TVS HARVEST-ALIGNED:

EMI: ₹1,000 / mo  
Impact: Buffer cushion for Rabi season

# CONVERSATIONAL GENAI: TVS SAHAYAK & RURAL UI/UX

## TVS SAHAYAK // GOOGLE GEMINI COPILOT

### ► Powered by Google Gemini 2.5 Flash:

Connects via server-side proxy (/api/assistant) with auto-fallback to Gemini 1.5 Flash. Fast, low-latency rural dialogue.

### ► Pre-Prompted Domain Grounding:

System instruction explicitly trained on TVS Credit Kisan Loan Hub products, Sentinel-2 NDVI spectral rules, and RBI KFS guidelines.

### ► Multilingual Audio & Text (Hindi/English):

Web Speech API Speech-to-Text allows farmers to speak naturally; Text-to-Speech audio read-out empowers low-literacy borrowers.

### ► Multi-Turn Conversation Memory:

Remembers previous applicant queries, translates complex APR & SHAP terms into farming analogies, and provides toll-free 1800-425-4500 escalation.

## UI/UX DIFFERENTIATION FOR RURAL BHARAT

### ★ Pulse 2026 Neo-Brutalism:

3px solid black borders, flat hard shadows, and high-contrast canary yellow/coral accents engineered for high visibility under harsh direct sunlight.

### ★ Zero-Connectivity Offline PWA:

Built with Serwist Turbopack service worker precaching 48 assets. IndexedDB (idb-keyval) stores boundary drafts offline and syncs on reconnect.

### ★ Low-Data Mode Bandwidth Saver:

One-tap toggle disables heavy satellite map tiles and renders lightweight vectors, saving 85% of mobile data for rural users.

### ★ Web Push Notifications (VAPID):

Zero-cost browser push alerts remind farmers 7 days prior to harvest EMIs and notify approval without expensive SMS gateway overheads.

# ANNEXURE 1: CODE ARCHITECTURE, SCHEMAS & STANDARDS

## NEXT.JS 16 & POSTGIS SCHEMA

- Next.js 16.3 App Router with strict Server/Client component boundaries.
- PostGIS geometry schema: plots.geom geometry(Polygon,4326) with ST\_Area and ST\_Distance spatial indexes.
- Drizzle ORM dual-mode: production connection pooling with in-memory fallback for offline zero-downtime execution.

## SECURITY, RATE LIMITING & ENCRYPTION

- Zod contract validation across all 15 API routes, preventing injection attacks.
- Sliding-window rate limiter in proxy.ts (30 req/min for Gemini AI, 5 OTPs/10min).
- Hardware AES-256-GCM encryption for all sensitive PII (Aadhaar, phones, survey numbers).
- 4-Tier RBAC: farmer, field\_officer, credit\_officer, and admin roles strictly verified.

## TESTING RIGOR & CI/CD PIPELINE

- Vitest test suite validating loan amortization, APR math, and KFS schedules.
- Pytest test suite validating FastAPI satellite feature extraction & SHAP attribution.
- Playwright E2E browser tests automating complete farmer apply-to-decision journeys.
- GitHub Actions CI pipeline executing type-checking, linting, and automated builds on every commit.

## ANNEXURE 2: RBI DIGITAL LENDING & DPDP ACT 2023 SOVEREIGNTY

### RESERVE BANK OF INDIA (RBI) DIGITAL LENDING

► **Statutory Key Fact Statement (KFS):**

Mandatory transparent statement generated upfront before loan execution (/kfs/[id]), detailing APR, 1.5% processing fee, 18% GST, stamp duty, and net disbursed cash.

► **Mandatory 3-Day Cooling-Off Period:**

Borrowers can exit the loan contract within 72 hours of disbursement without any prepayment penalty by returning principal plus proportionate daily interest.

► **100% Direct Account Disbursal:**

Loan proceeds are transferred strictly into the borrower's verified bank account; zero third-party pass-through wallets or prepaid cards permitted.

► **Statutory Grievance Redressal:**

Direct in-app escalation pathways to the TVS Credit Grievance Redressal Officer and RBI Integrated Ombudsman.

### DIGITAL PERSONAL DATA PROTECTION (DPDP) ACT 2023

★ **Consent Management Portal (/consent):**

Farmers inspect all individual consents granted (eKYC, satellite telemetry, mandi financial analysis) with granular one-tap withdrawal capability.

★ **Section 12 Right to Erasure:**

Permanent data scrubbing button allowing borrowers to permanently delete field boundaries, OCR documents, and contact records from TVS systems.

★ **Purpose-Bound Data Retention:**

Satellite plot telemetry retained only for the duration of the loan cycle plus one harvest season, in strict compliance with DPDP rules.

★ **Immutable Audit Trail (/api/audit):**

Every staff view, credit committee decision, and consent modification logged with client IP, timestamp, and SHA-256 tamper-evident hashing.

## ANNEXURE 3: EARLY WARNING RADAR, SCALING & GREEN FINANCING

### PHASE 8: PORTFOLIO RISK RADAR & AUTOMATED SURVEILLANCE

- ★ Daily Upstash QStash crons continuously inspect Sentinel-2 NDVI across all active loan parcels.
- ★ Triggers alerts if vegetation vigor drops >15% (pest/disease) or NASA weather anomaly exceeds -30% (drought).
- ★ Proactive credit mitigation: automated vernacular SMS advisory, 60-day EMI moratorium offer, and geo-tagged field officer dispatch.

### PHASE 9: ENTERPRISE CLOUD HARDENING & REDIS CACHING

- ★ Upstash Redis distributed caching: 24h satellite NDVI cache & 3h weather forecast cache ensuring <100ms response times.
- ★ Multi-region deployment: Next.js edge runtime on Vercel Mumbai (bom1) and FastAPI geo-ml microservice on Railway/Render.
- ★ Sentry APM distributed tracing and structured JSON error logging for bank-grade reliability.

### PHASE 10: INDIA DIGITAL AGRI-STACK & ESG GREEN FINANCING

- ★ Direct integration with Government of India AgriStack & Farmer Registry API for instant digital land ownership validation.
- ★ Account Aggregator (AA) integration (Sahamati) to verify non-farm dairy and allied rural cashflows.
- ★ Green Agri-Credit: Satellite biomass tracking unlocks 50 bps interest rate rebates for farmers adopting drip irrigation and solar water pumps (PM-KUSUM).